

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: SUB-STRAND 2.4 AREA OF POLYGONS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can we calculate the exact area of any polygon-shaped surface, even when we cannot measure its height, so that builders, farmers and designers in our community never order too much or too little material? Use what you learned in the Kericho community hall courtyard unit to answer each prompt. Write in full sentences, show every step of your working, include units, and give reasons for your choices. Round areas to 1 decimal place unless told otherwise. You may use a calculator and your Courtyard Model.

Part 1: The Courtyard Puzzle and the Sine-Area Formula

A contractor in Kericho County must order tiles for a community hall courtyard. Some triangular sections have two known side lengths and the angle between them, but no height can be measured and there is no right angle. (a) Explain why $A = \frac{1}{2} \times \text{base} \times \text{height}$ cannot be used directly on these triangles. (b) Using a right-angled sub-triangle, show how the height can be written using side b and the included angle C. (c) Substitute this into $A = \frac{1}{2} \times \text{base} \times \text{height}$ to get a formula for the area. (d) Explain how this formula helps the contractor with Triangle A and Triangle B of the courtyard.

Write your answer here:

Part 2: Triangles with Three Sides Only (Heron's Formula)

Mwangi, a farmer near Eldoret in the Rift Valley, has a triangular maize plot with sides 40 m, 55 m and 63 m. He cannot measure any angle or any height. (a) Explain why $\frac{1}{2}ab \sin C$ is not the best choice for his plot. (b) Write down Heron's Formula and say what s stands for. (c) Calculate the area of the plot, showing each step. (d) A classmate says the answer will change if you call the 63 m side a instead of c . Do you agree? Explain using what you found in class.

Write your answer here:

Part 3: The Parallelogram Stage – Choosing the Right Method

The stage in the Kericho courtyard is a parallelogram with adjacent sides 6 m and 4 m and an included angle of 65° . (a) Write the formula you would use and explain why it is the practical choice here. (b) Calculate the area of the stage. (c) Explain how this formula is connected to the triangle formula $\frac{1}{2}ab \sin C$. (d) A kanga patch is a rhombus whose two diagonals can both be measured. Which formula would you choose for it, and why? (e) The contractor needs whole tiles. Should the number of tiles be rounded up or down? Why?

Write your answer here:

Part 4: Regular Pentagons and Hexagons from Side Length Alone

(a) A njugu (groundnut) vendor at Kericho market uses a regular pentagonal tray with side length s . Explain, step by step, how dividing the pentagon into congruent triangles leads to $A = 5s^2/(4 \tan 36^\circ)$. Include the angles in one triangle and say why $\tan 36^\circ$ appears. (b) Use the formula to find the area of a tray with $s = 9$ cm. (c) The hexagonal border tiles of the courtyard are regular hexagons with known side length s . Explain how to derive their area formula from the area of an equilateral triangle and say why no height measurement is needed.

Write your answer here:

Part 5: Putting It All Together for the Kericho Courtyard

Your final task is to explain the answer to the driving question. (a) For each courtyard section, say which formula you would use and what measurements the contractor needs: (i) a triangle with two sides and the included angle; (ii) a triangle with three sides; (iii) the parallelogram stage; (iv) a hexagonal border tile. (b) Explain why the Lesson 1 approach of base $\times$ height only could not give the total area. (c) Explain how knowing every section's area helps the contractor avoid ordering too much or too little material.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Choosing and justifying the correct formula	Selects the correct formula for every shape and explains clearly, using the available measurements, why it is the practical choice when no height can be measured.	Selects the correct formula for most shapes and gives a reasonable but incomplete reason.	Chooses formulas incorrectly or without giving a reason.
Derivations and links between formulas	Shows correctly how $h = b \sin C$ leads to $\frac{1}{2}ab \sin C$, and links the parallelogram, equilateral triangle, hexagon and pentagon formulas to the triangle formulas with clear steps and angles.	Gives most steps of the derivations, with a small gap in the angles or in the link between formulas.	States formulas without showing how they are derived or connected.
Accurate calculation	Calculations are correct and fully shown (Heron's $\approx 1,087.7 \text{ m}^2$, stage $\approx 21.75 \text{ m}^2$, pentagon $\approx 139.4 \text{ cm}^2$), with units and sensible rounding.	Method is correct but with a minor arithmetic, rounding or unit error.	Method is incorrect or the working is not shown.
Connecting to the Kericho courtyard and driving question	Clearly explains how the formulas let the contractor find every area from measurable data and avoid over- or under-ordering, including rounding up tile counts.	Links the formulas to the courtyard but gives limited detail on ordering material.	Gives little or no link to the courtyard or the driving question.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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